



ANDERSON JUNIOR COLLEGE
2013 JC2 PRELIMINARY EXAMINATIONS

CHEMISTRY

9647/01

Higher 2

23 September 2013

Paper 1 Multiple Choice

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

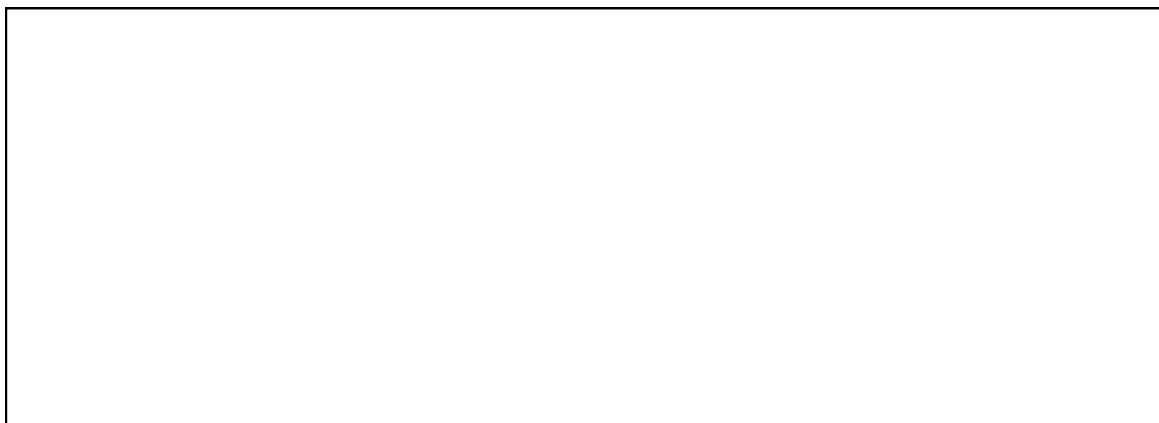
Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

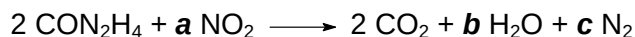


This document consists of **18** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

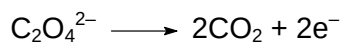
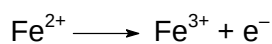
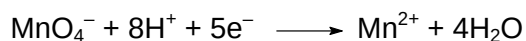
- 1 Granular urea, CON_2H_4 , can be used to remove NO_2 from the flue gases of power stations, converting it into harmless nitrogen.



What are the values of **a**, **b** and **c** in a balanced equation?

	a	b	c
A	$\frac{3}{2}$	2	$\frac{5}{4}$
B	2	4	3
C	3	4	$\frac{7}{2}$
D	3	4	3

- 2 Consider the following half-equations.



What volume of 0.02 mol dm^{-3} KMnO_4 is required to oxidise 20 cm^3 of an acidified solution of 0.05 mol dm^{-3} FeC_2O_4 ?

- A** 10 cm^3 **B** 20 cm^3 **C** 30 cm^3 **D** 50 cm^3
- 3 *Use of the Data Booklet is relevant to this question.*

On 9th October 2006, a team of scientists from America and Russia reported the discovery of element 118, the heaviest element yet discovered. The atoms had a relative atomic mass of 294. The element has provisionally been named Ununoctium and given the symbol Uuo.

Which statement about this element and its compound is correct?

- A** It has 294 neutrons.
B It forms a square planar tetroxide UuoO_4 .
C It has a higher first ionisation energy than that of $_{86}\text{Rn}$.
D It has two more occupied electron shells than a Cs^+ ion.

- 4 Two conversions are outlined below.

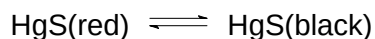


What similar feature do these two conversions have?

- A** a lone pair of electrons in the product
B change in oxidation state of an element
C decrease in bond angle of the species involved
D disappearance of a π bond
- 5 Two identical bulbs at the same temperature contain ideal gases **D** and **E** separately. The density of gas **D** is twice that of gas **E** and the molecular mass of gas **D** is half that of gas **E**.

What is the ratio of the pressure of gas **D** to that of gas **E**?

- A** 1 : 2 **B** 1 : 1 **C** 2 : 1 **D** 4 : 1
- 6 Red mercury(II) sulfide is used as a red pigment which is known to darken and this has been ascribed to the equilibrium below.



At temperatures below 400 °C, red HgS is the stable form of HgS.

Data for each form of HgS are given in the table.

	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$	$S^\ominus / \text{J mol}^{-1} \text{K}^{-1}$
red	-58.2	82.4
black	-41.0	108.0

What is the expression for $\Delta G^\ominus$, in J mol^{-1} , for the formation of black HgS from red HgS at 450 °C?

- A** $\Delta G^\ominus = +17.2 - 723(+25.6)$
B $\Delta G^\ominus = +17.2 - 450(-25.6)$
C $\Delta G^\ominus = +17200 - 450(-25.6)$
D $\Delta G^\ominus = +17200 - 723(+25.6)$

- 7 Polymerisation is a process in which large number of small molecules (monomers) are joined together to form long-chain molecules.

Considering that polymerisation is a spontaneous reaction, what are the correct signs of ΔS and ΔH for the process?

	ΔS	ΔH
A	–	–
B	–	+
C	+	–
D	+	+

- 8 *Use of the Data Booklet is relevant to this question.*

Lithium cells are used in digital watches, hearing aids and mobile phones.

They consist of lithium and manganese(IV) oxide in non-aqueous pastes enclosed between sheets of stainless steel.

The electrode reactions are as follows.

negative electrode: $\text{Li(s)} \longrightarrow \text{Li}^{\text{+}}\text{(s)} + \text{e}^{-}$

positive electrode: $\text{Li}^{\text{+}}\text{(s)} + \text{MnO}_2\text{(s)} + \text{e}^{-} \longrightarrow \text{LiMnO}_2\text{(s)}$

The capacity of a cell is given in amp hours (Ah), which is the amount of constant current that can be drawn from the cell in one hour before it becomes discharged.

Which expression will calculate the minimum mass of lithium required to make a cell with a capacity of 1.8 Ah?

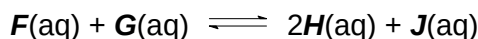
A $\frac{1.8 \times 3600 \times 6.9}{96500} \text{ g}$

B $\frac{1.8 \times 3600 \times 6.9}{1.6 \times 10^{-19}} \text{ g}$

C $\frac{1.8 \times 96500 \times 6.9}{3600} \text{ g}$

D $\frac{1.8 \times 1.6 \times 10^{-19}}{6.9 \times 3600} \text{ g}$

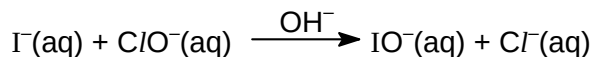
- 9 An equilibrium can be represented by the following equation.



In a certain 1 dm³ mixture, the equilibrium concentration of **G** is 10 mol dm⁻³.

What will be the new equilibrium concentration of **G** if 5 mol of pure **G** is dissolved in the mixture?

- A** 15 mol dm⁻³
B between 10 mol dm⁻³ and 15 mol dm⁻³
C 10 mol dm⁻³
D between 5 mol dm⁻³ and 10 mol dm⁻³
- 10 Which one of the following statements is correct about a reaction for which the equilibrium constant is independent of temperature?
- A** The enthalpy change is zero.
B Its rate constants do not vary with temperature.
C There are equal number of moles of reactants and products.
D The forward and reverse reactions are of the same order.
- 11 The kinetics of the reaction below was studied using the initial rates method.



Experiment	Initial $[I^{-}]$ / mol dm ⁻³	Initial $[ClO^{-}]$ / mol dm ⁻³	Initial $[OH^{-}]$ / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.0013	0.012	0.100	9.4×10^{-3}
2	0.0026	0.012	0.100	18.7×10^{-3}
3	0.0013	0.018	0.100	14.0×10^{-3}
4	0.0026	0.018	0.050	56.1×10^{-3}

What is the overall order of the reaction?

- A** 1 **B** 2 **C** 3 **D** 4
- 12 The decomposition of hydrogen peroxide in aqueous solution is a first order process.

If $\frac{2}{3}$ of the hydrogen peroxide in a solution of known concentration decomposes in 52 minutes at a certain temperature, what is the time taken for the same concentration of hydrogen peroxide to decompose to $\frac{1}{8}$ of its initial concentration?

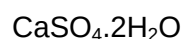
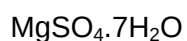
- A** 267 min **B** 156 min **C** 104 min **D** 98 min

- 13** The oxide and chloride of an element **K** are separately mixed with water. The two resulting solutions have the same effect on litmus.

Which of the following can be element **K**?

- A** sodium
- B** aluminium
- C** silicon
- D** phosphorus

- 14** The sulfates of Group II elements exist in crystalline form as follows.



What is the reason for this trend in hydration?

- A** The atomic radius of the elements increases down the group.
 - B** The ionic character of these sulfates increases.
 - C** The ionisation energy of the elements decreases down the group.
 - D** The radius of the cations increases down the group.
- 15** The following two experiments are carried out with anhydrous potassium chloride and observations *X* and *Y* are made at the end of each experiment.

Concentrated sulfuric acid is added to the potassium chloride and the fumes produced are bubbled into aqueous potassium bromide solution – observation *X*.

The potassium chloride is dissolved in concentrated aqueous ammonia and this is then added to aqueous silver nitrate – observation *Y*.

What are the observations *X* and *Y*?

	<i>X</i>	<i>Y</i>
A	brown solution	colourless solution
B	brown solution	white precipitate
C	colourless solution	colourless solution
D	colourless solution	white precipitate

- 16 Sulfur is converted to SF_6 by fluorine, to SCl_2 by chlorine and to S_2Br_2 by bromine.

Which trend does this information best provide evidence for?

- A the trend in bond energy: $\text{F}_2 < \text{Cl}_2 > \text{Br}_2$
 - B the trend in oxidising ability: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2$
 - C the trend in electronegativity: $\text{F} > \text{Cl} > \text{Br}$
 - D the trend in first ionisation energy: $\text{F} > \text{Cl} > \text{Br}$
- 17 Which statement correctly **defines** a transition element?
- A Transition elements have partially filled d-orbitals.
 - B Transition elements form many coloured compounds.
 - C Transition elements or their compounds are widely used as catalysts.
 - D Transition elements exhibit more than one oxidation state in their compounds.
- 18 When drops of $\text{NH}_3(\text{aq})$ are added to $\text{Cu}(\text{NO}_3)_2(\text{aq})$, a pale blue precipitate is formed. This precipitate dissolves when an excess of $\text{NH}_3(\text{aq})$ is added, forming a deep blue solution.

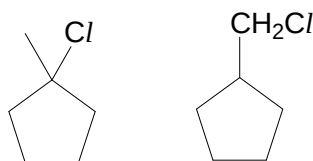
Which process does **not** occur in this sequence?

- A acid–base reaction
 - B formation of a complex ion
 - C ligand exchange
 - D reduction of Cu^{2+} ions
- 19 Covalent bonds are formed by orbitals overlap. The shape of unsaturated hydrocarbon molecules can be explained in terms of hybridisation of orbitals.

Which bond is **not** present in $\text{HC}\equiv\text{CCH}_2\text{CH}=\text{CH}_2$?

- A a π bond formed by $2\text{sp}-2\text{sp}$ overlap
- B a σ bond formed by $1\text{s}-2\text{sp}^2$ overlap
- C a σ bond formed by $2\text{sp}^2-2\text{sp}^2$ overlap
- D a σ bond formed by $2\text{sp}-2\text{sp}^3$ overlap

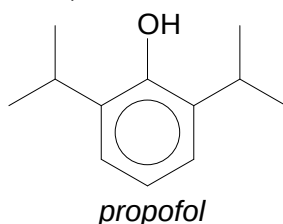
- 20 Methylcyclopentane can react with chlorine via free radical substitution to produce a mixture of four monochlorinated products, two of which are shown below.



For methylcyclopentane, the order of reactivity of tertiary and primary hydrogen atoms follows a 5 : 1 ratio.

What is the likely ratio of tertiary RCl : primary RCl formed?

- A 1 : 15
 B 1 : 3
 C 3 : 5
 D 5 : 3
- 21 Which of these compounds, on complete combustion of one mole, will release the greatest amount of energy?
- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 B $\text{CH}_3\text{CH}_2\text{CH}_3$
 C $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
 D CH_3COCH_3
- 22 The diagram shows the structure of *propofol*, a short-acting intravenous anaesthetic used for both adults and children, and also in veterinary medicine.



When reacted with a very dilute solution of Cl_2 , a chlorine atom may substitute for a hydrogen atom on the benzene ring but **not** for a hydrogen atom on the alkyl branches or in the $-\text{OH}$ group.

Given that any number of the benzene hydrogen may be substituted, how many possible products of the reaction are there?

- A 1 B 3 C 5 D 7

- 23 Four isomers of $C_3H_6OCl_2$ are separately subjected to hydrolysis using aqueous sodium hydroxide.

Which isomer produces an organic product with a different molecular formula from the other three isomers?

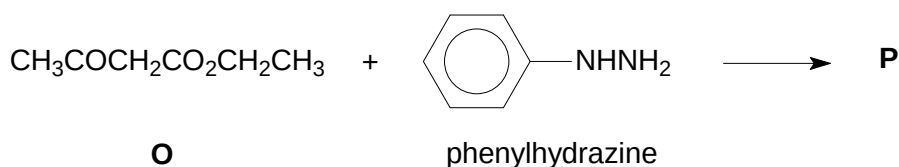
- A $CH_3CCl_2CH_2OH$
- B $CH_2ClCHClCH_2OH$
- C $CH_2ClCH_2CH(OH)Cl$
- D $CHCl_2CH_2CH_2OH$

- 24 An organic compound **N** reacts with sodium to produce an organic ion with a charge of -3 . **N** reacts with $NaOH(aq)$ to produce an organic ion with a charge of -1 .

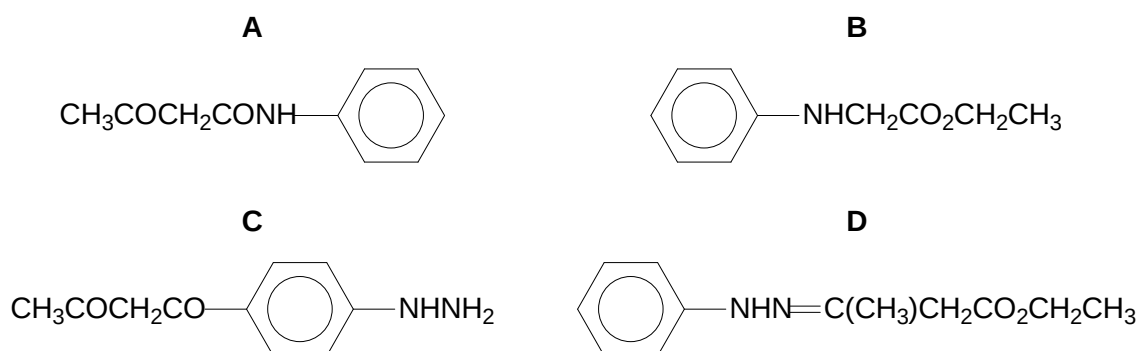
What could be the structural formula of **N**?

- A $HO_2CCH(OH)CH_2CO_2H$
- B $HO_2CCH(OH)CH_2CHO$
- C $HOCH_2CH(OH)CH_2CO_2H$
- D $HOCH_2COCH_2CHO$

- 25 The first stage in the synthesis of *Antipyrine*, a drug used in reducing fever, is the reaction between compound **O** and phenylhydrazine.



What is the product **P** of this first stage?

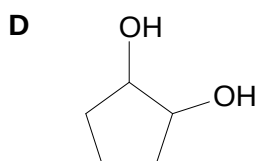


26 Use of the table on page 8 of the Data Booklet is relevant to this question.

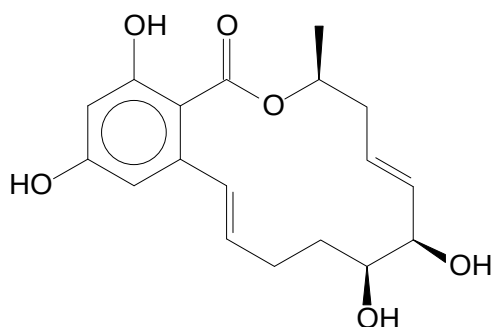
An organic liquid **Q** with molecular formula $C_5H_{10}O_2$, shows a broad absorption at $3100\text{--}3500\text{cm}^{-1}$ in the infra-red spectrum. When **Q** reacts with acidified sodium dichromate(VI) solution under mild conditions, a liquid can be distilled from the reaction mixture. This liquid gives a brick-red precipitate on warming with Fehling's reagent.

Which of the following is the likely identity of compound **Q**?

- A $CH_3(CH_2)_3CO_2H$
 B $CH_3CH(OH)CH_2COCH_3$
 C $HOCH_2CH_2COCH_2CH_3$



27 Research has shown that *Aigialomycin D*, a fungal metabolite, is capable of inhibiting crucial enzymes related to cancers.



Aigialomycin D

Which of the following observation is **incorrect**?

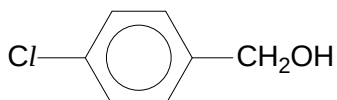
- A It reacts with hot, acidified $K_2Cr_2O_7$ to give a diketone.
 B It reacts with hot, alkaline aqueous iodine to give a yellow solid.
 C It reacts with $Br_2(aq)$ to incorporate up to 6 atoms of bromine in each molecule.
 D It reacts with cold, dilute acidified $KMnO_4$ to give a product that has 7 chiral centres.

28 An organic compound **R** has the following properties:

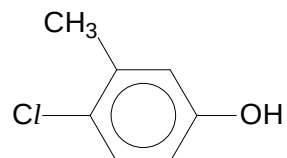
- (i) It has the molecular formula C_7H_7OCl .
- (ii) When $Br_2(aq)$ is added to **R**, a white solid, with M_r of 300.3, is formed.
- (iii) Warming **R** with $AgNO_3(aq)$ gives a white solid, which is soluble in $NH_3(aq)$.

What is the likely identity of compound **R**?

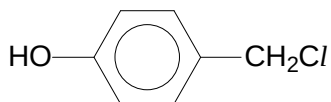
A



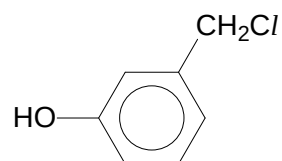
B



C

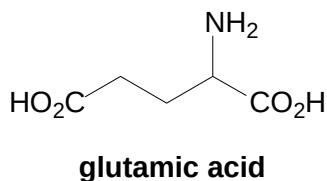


D



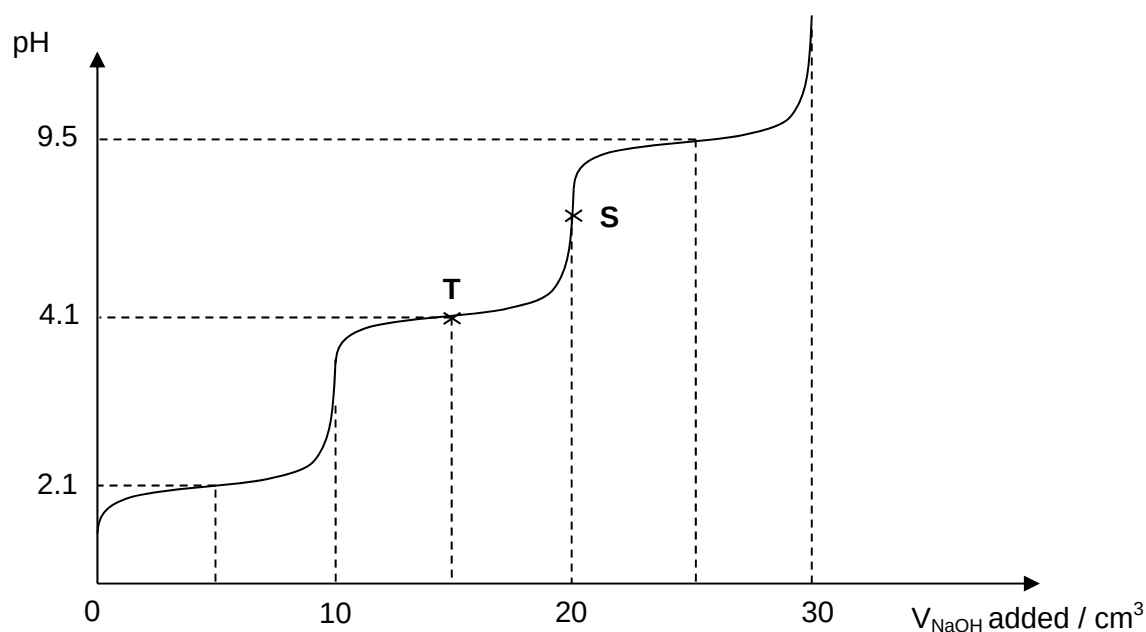
The following information is relevant to questions **29** and **30**.

Soy sauce is produced by the fermentation of soy beans by the mould *Aspergillus oryzae*. The distinctive salty taste of the sauce is due to salts of glutamic acid formed during fermentation.

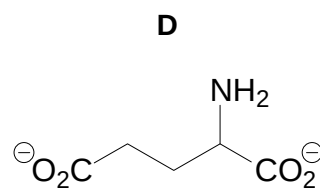
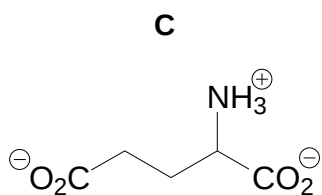
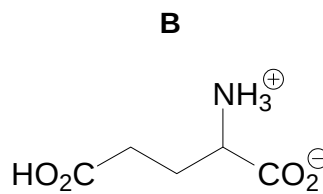
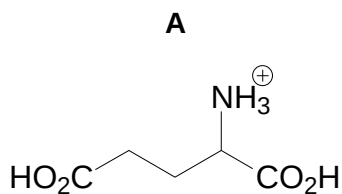


There are three pK_a values associated with glutamic acid: 2.1, 4.1 and 9.5.

The pH–volume curve obtained when 30 cm^3 of NaOH is added to 10 cm^3 of the protonated form of glutamic acid of the same concentration is given below.



29 Which of the following is the major species present at point **S**?



- 30** Electrophoresis is a technique of separating and identifying amino acids. A solution of amino acids is absorbed into the paper that is moistened with a buffer solution and stretched between two electrodes. Positively charged species move towards the cathode, negatively charged towards the anode.

At point **T**, there are two forms of glutamic acid present in equal amount.

With a buffer at pH 4.1, what will happen to the two forms of glutamic acid when an electrophoresis experiment was carried out?

- A** Both migrate to the anode.
- B** Both migrate to the cathode.
- C** One remains stationary while the other migrates to the anode.
- D** One remains stationary while the other migrates to the cathode.

Section B

For each of the question in this section, one or more of the three numbered statements **1** to **3** may be correct.

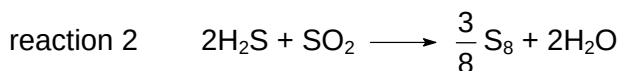
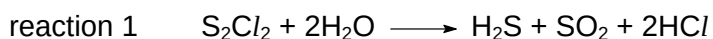
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

31 The hydrolysis of S_2Cl_2 proceeds by two reactions.



Which of the following correctly describe the reactions?

- 1** An acidic solution is formed in reaction 1.
- 2** Reverse disproportionation occurs in reaction 2.
- 3** The oxidation state of chlorine and oxygen remain unchanged in both reactions.

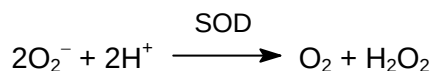
32 Gaseous particle **U** has an atomic number n and a charge of $+1$.

Gaseous particle **V** has an atomic number of $(n + 1)$ and is isoelectronic with **U**.

Which of the following statements is always true?

- 1** **V** has a smaller radius than **U**.
- 2** **V** is deflected as much as **U** when passed through an electric field.
- 3** **V** releases less energy than **U** when an electron is added to each particle.

- 33** Superoxide dismutase (SOD) is an example of a copper-containing enzyme. It aids in the reaction of the strongly oxidising superoxide ion, O_2^- , to oxygen and hydrogen peroxide, thereby protecting the body from harm.



The activity of SOD hinges on the active site Cu^{2+} ion which can be reduced to Cu^+ .

The $E^\ominus$ data, where the oxidised and reduced forms of the enzyme are represented as SOD-Cu^{2+} and SOD-Cu^+ , are given in the table below.

half-equations	$E^\ominus / \text{V}$
$\text{O}_2 + \text{e}^- \rightleftharpoons \text{O}_2^-$	-0.33
$\text{O}_2^- + \text{e}^- + 2\text{H}^+ \rightleftharpoons \text{H}_2\text{O}_2$	+0.89
$\text{SOD-Cu}^{2+} + \text{e}^- \rightleftharpoons \text{SOD-Cu}^+$	+0.42

Which of the following statements are true?

- 1 Disproportionation of superoxide ion, O_2^- , occurs.
 - 2 The reaction of superoxide ion, O_2^- , has high activation energy.
 - 3 The first step of the enzyme-catalysed reaction mechanism is more spontaneous than the second.
- 34** Carbon and silicon each form a tetrachloride. CCl_4 has no reaction with water; SiCl_4 reacts violently with water.

Which statement is a correct explanation of this?

- 1 Si-Cl bond is weaker than C-Cl bond.
- 2 Carbon does not have low-lying empty d-orbitals.
- 3 The four large chlorine atoms shield the small carbon atom from the water molecules.

The responses **A** to **D** should be selected on the basis of

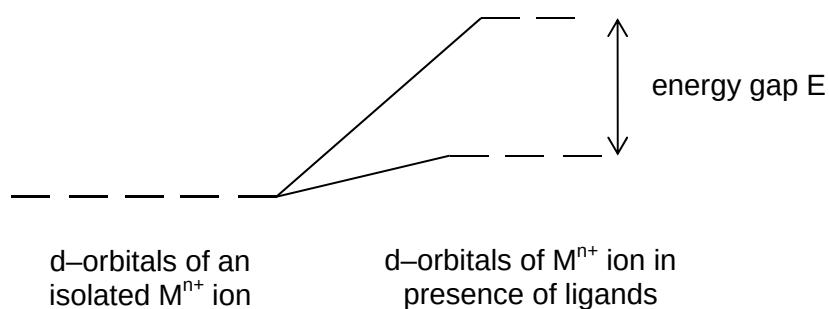
A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

35 *Use of the Data Booklet is relevant to this question.*

The arrangement of electrons in the d-orbitals depends on the spin states of complexes.

The following diagram shows how the d-orbitals are split in an octahedral environment.



In a 'high spin' state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

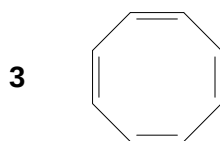
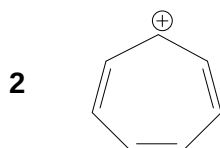
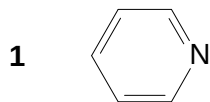
In a 'low spin' state, the lower energy d-orbitals are filled first, by pairing up if necessary, before the higher energy d-orbitals are used.

Which of the following divalent cations are able to exhibit both spin states in the presence of different ligands?

- 1** Co^{2+}
- 2** Mn^{2+}
- 3** Ni^{2+}

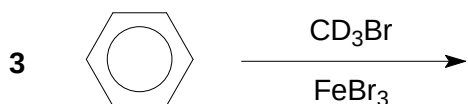
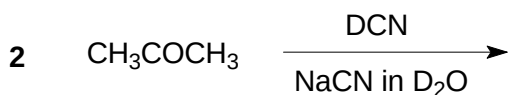
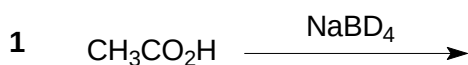
- 36** In organic chemistry, Hückel's rule is used to predict whether a planar monocyclic, conjugated molecule will have aromatic properties. It was found that aromatic molecule has a total of $(4n + 2) \pi$ electrons, where n is an integer ($n = 0, 1, 2, 3, \dots$).

Which of the following monocyclic molecule / ion exhibit aromaticity?

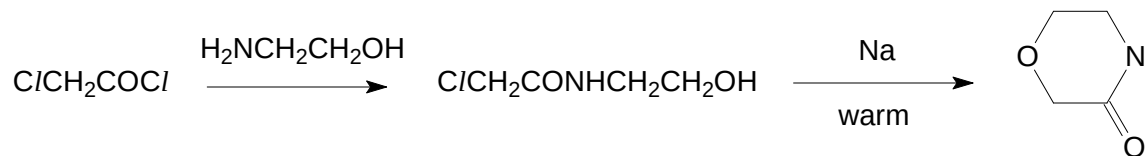


- 37** Deuterium, D, is the ^2_1H isotope of hydrogen.

Which reactions yield an organic compound incorporating deuterium, D?



- 38** Below is a reaction scheme involving chloroacetyl chloride.



What are the types of reaction shown in this scheme?

- 1** redox
- 2** nucleophilic substitution
- 3** acid–base reaction

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

39 The compound **X**, $\text{CH}_3\text{COCH}_2\text{CHO}$, was reacted in separate experiments with

- (i) 2,4–dinitrophenylhydrazine reagent;
- (ii) Tollen's reagent;
- (iii) alkaline aqueous iodine.

Which statements about these reactions are correct?

- 1** One mole of **X** could react with two moles of 2,4–dinitrophenylhydrazine in (i).
- 2** One mole of **X** could form two moles of Ag(s) in (ii).
- 3** One mole of **X** could form two moles of $\text{CHI}_3\text{(s)}$ in (iii).

40 Equal amounts of two organic compounds, **Y** and **Z**, were added to water and the pH values of both solutions were determined. It was found that the pH of the aqueous solution of **Y** is higher.

Which pairs of compounds could be **Y** and **Z**?

	Y	Z
1	$(\text{CH}_3)_2\text{NH}$	CH_3NH_2
2	$\text{CH}_3\text{COCH}_2\text{NH}_2$	$\text{CH}_3\text{CH}_2\text{CONH}_2$
3	$\text{C}_6\text{H}_5\text{O}^-\text{Na}^+$	$\text{C}_6\text{H}_5\text{CO}_2^-\text{Na}^+$